PROJECT PRESENTATION

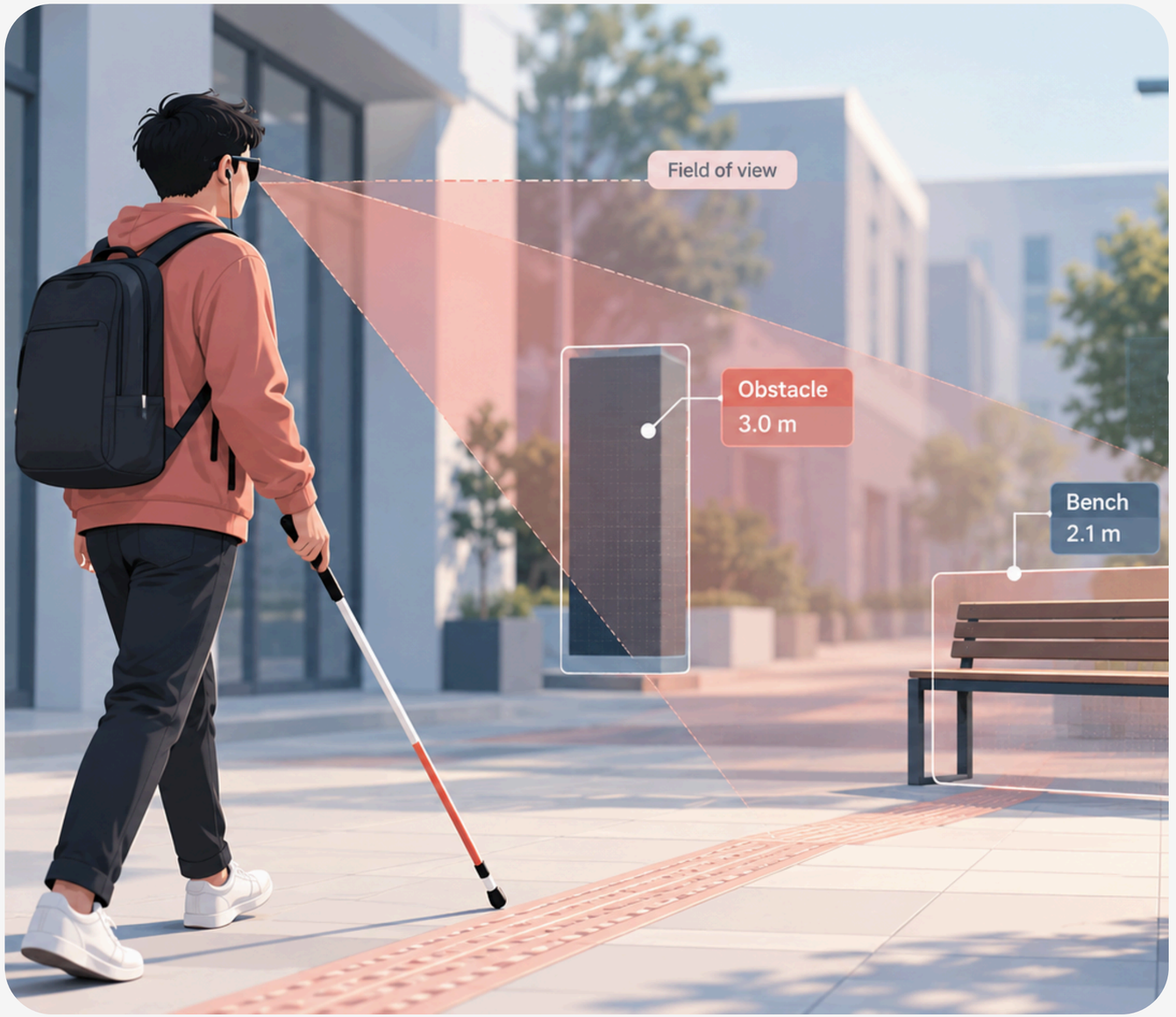
# Echo Vision

An AI-powered assistive perception system for visually impaired individuals

Unity-based digital twin approach


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# Introduction

Echo Vision is an AI-powered assistive perception concept that aims to help visually impaired individuals understand their surroundings through intelligent environmental sensing and meaningful information delivery.

The system will combine virtual environment simulation, computer vision concepts and spatial perception to turn raw environmental information into clear guidance.

Development follows a Unity-based digital twin approach. Perception capabilities will first be designed and refined in a virtual environment, before any physical hardware is introduced.

# Problem Statement

Visually impaired individuals find it difficult to identify objects along their path, to judge how far obstacles are, and to move through unfamiliar places without assistance. Conventional mobility aids support physical navigation, but offer limited contextual information about what lies ahead.

- Obstacle awareness
- Distance awareness
- Environmental understanding
- Unfamiliar environments

**Problem definition.** To develop an intelligent assistive perception system that interprets environmental information and provides meaningful feedback to visually impaired individuals.

# Objectives

01 Build a Unity-based digital twin of real-world surroundings

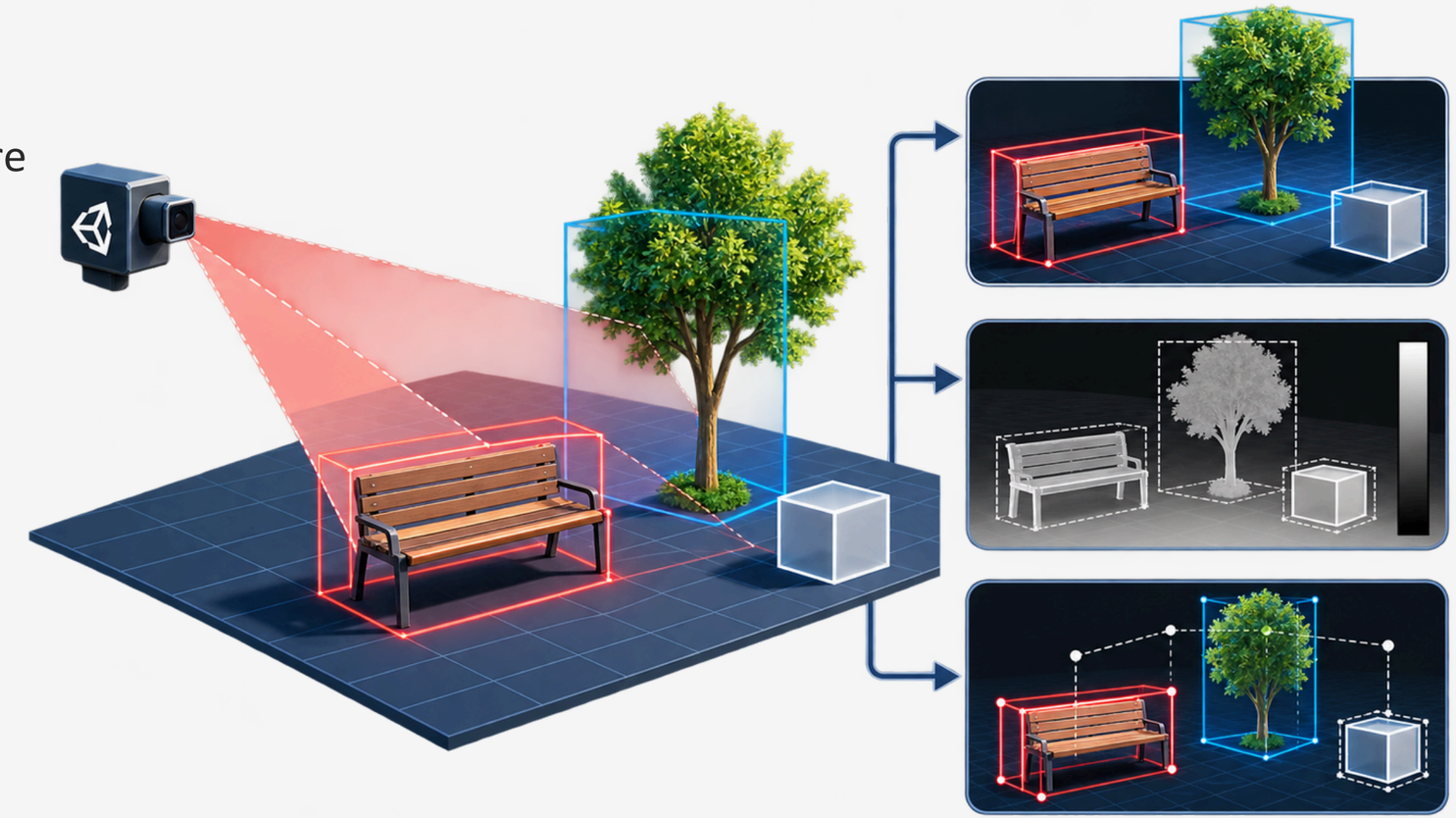
02 Establish a virtual camera-based capture system

03 Detect ground surfaces and potential obstacles

04 Identify objects and estimate their distances

05 Develop a structured environmental perception pipeline

06 Lay the foundation for audio guidance and hardware integration



# Proposed Solution

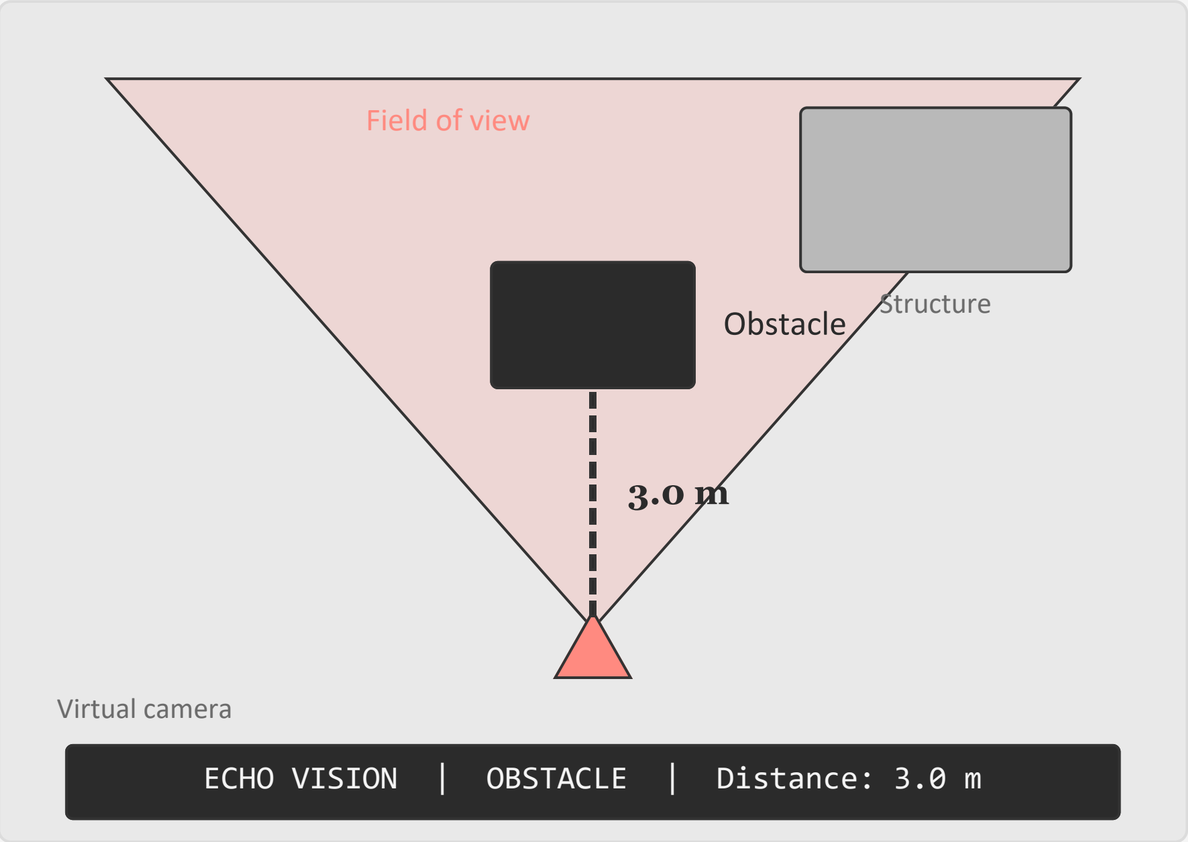
Echo Vision proposes an integrated pipeline that captures the surrounding environment, analyses it, and converts the result into understandable guidance. A virtual camera will observe the Unity scene, and a custom perception component will process what it sees.



# Digital Twin Methodology

A digital twin is a virtual representation of a physical environment that can be modelled, observed and evaluated.

Unity will serve as this virtual environment, giving the perception pipeline controlled, repeatable scenes with flexible object placement and less early dependence on hardware.




Concept illustration

# System Architecture

LAYER 01

### Environmental layer

Represents the virtual surroundings: ground surfaces, structures, obstacles and scene objects.

LAYER 02

### Capture layer

Captures the rendered environment through the virtual camera feed, ECHO_VISION_CameraFeed (256 × 256).

LAYER 03

### Virtual sensing layer

ECHO_VISION_RangeSensor acquires the rendered output and runs the scanning logic.

LAYER 04

### Perception layer

Ground detection, obstacle detection, object identification and distance estimation.

LAYER 05

### Information layer

Organises perception results into structured information for higher-level functions.

LAYER 06

### Assistive output layer

Translates environmental information into understandable audio guidance.

# Perception Workflow

The Unity camera will render the virtual scene into a 256 × 256 camera feed. A custom sensing component, ECHO_VISION_RangeSensor, will acquire each rendered frame through the RenderPipelineManager.endCamera event and run the scanning logic.

For every detection, the system will report what was found, whether it is ground or an obstacle, which object it belongs to, and its estimated distance.

PLANNED OUTPUT FORMAT

```
ECHO VISION  |  GROUND  |  Distance

ECHO VISION  |  OBSTACLE  |  Object  |
Distance
```

PLANNED AUDIO GUIDANCE

*Obstacle ahead, about three metres*

# Technology Stack

**Unity 3D**

Virtual environment and digital twin

**C#**

Sensing and perception programming

**Unity Camera**

Environmental image capture

**Rendering pipeline**

Camera-feed acquisition

**Computer vision and AI**

Environmental interpretation

**Audio technology**

Assistive information delivery

# Expected Impact

Echo Vision is intended to contribute to accessibility and assistive technology by giving users meaningful information about nearby objects, obstacles and spatial conditions.

01

**Independence**

Informed movement

02

**Awareness**

Spatial understanding

03

**Confidence**

Less uncertainty

04

**Inclusion**

Accessible technology

# Development Roadmap

The project will progress through six phases, moving from a simulated environment towards validation in real-world conditions.

PHASE 01

**Digital twin environment**

Develop and organise the virtual environment for assistive perception.

PHASE 02

**Environmental perception**

Refine ground and obstacle detection, object identification and distance estimation.

PHASE 03

**Contextual understanding**

Interpret multiple environmental elements together, in context.

PHASE 04

**Audio guidance**

Develop an audio feedback approach for perception information.

PHASE 05

**Hardware integration**

Explore integration with suitable physical sensing and computing components.

PHASE 06

**Real-world validation**

Evaluate reliability, usability and effectiveness in real environments.

# Conclusion

Echo Vision sets out to turn environmental information into meaningful guidance for visually impaired individuals. By developing perception in a Unity-based digital twin first, the project creates a controlled foundation for sensing, detection and distance estimation.

Audio guidance and physical hardware integration will then build on this tested foundation.

> *Helping people experience their surroundings beyond what their eyes can see.*

# Thank You

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